



《编译原理》

第1讲: 导言

何冬杰
重庆大学

课程概要

□理论课：周一3-4节、周三1-2节[1-4,6-11周]

●地点：DZ403, DZ216

□实验课：周日6-9节[5,7-8,10周]

●地点：DS1512

□学时：40+16=56节

□网站：<https://plc-cqu.github.io/teaching/index.html>

□主讲教师：何冬杰

●研究方向：编程语言与编译器

●先后在中科院计算所和UNSW的编译器组深造

□助教：熊锋

●用OCaml编写了课程实验全套工具：编译器、解释器、汇编器、链接器等

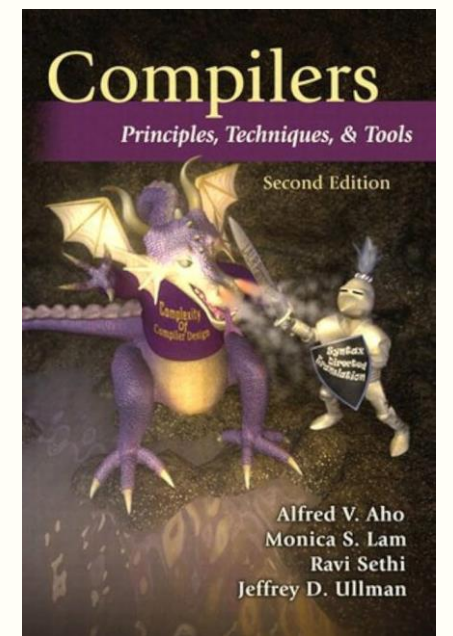
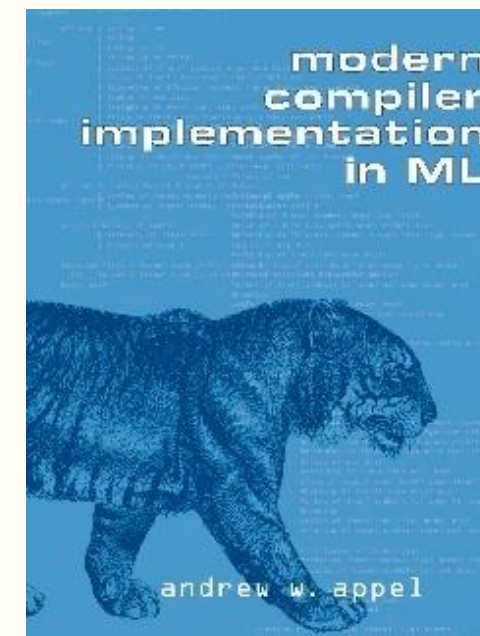


群聊：编译原理课程群



该二维码7天内(9月13日前)有效，重新进入将更新

Textbook



虎书

龙书

❑ In most cases, class materials should suffice

- Lecture slides posted after the lecture

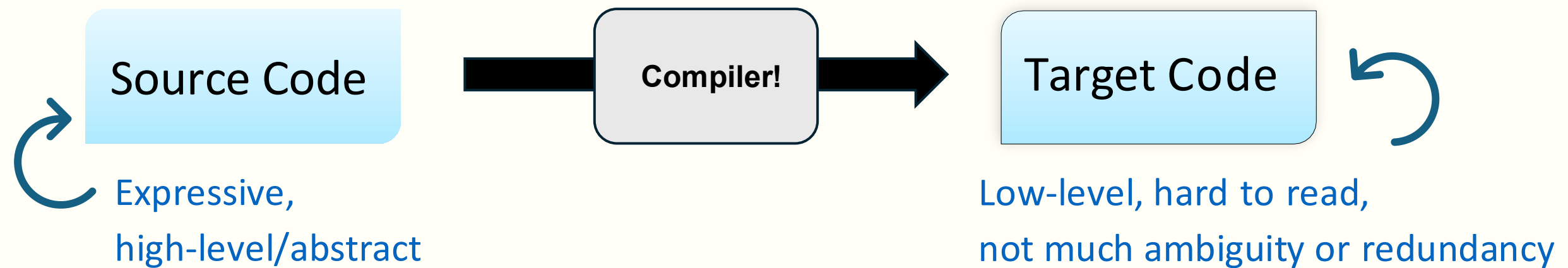
❑ Textbooks (*Recommended but not required*):

- Modern Compiler Implementation in ML by Andrew W. Appel
- Compilers: Principles, Techniques, and Tools (2nd Edition) by Alfred V. Aho, Monica S. Lam, Ravi Sethi, and Jeffrey D. Ullman

❑ Why Choose OCaml as the Programming environment?

- Ideal for writing compilers!
 - Designed to enable easy manipulation of abstract syntax trees
 - Type-safe, mostly pure, functional language with support for polymorphic (generic) algebraic datatypes, modules, and mutable state
- Less coding, more focus on compiler principles.

What is this course about?



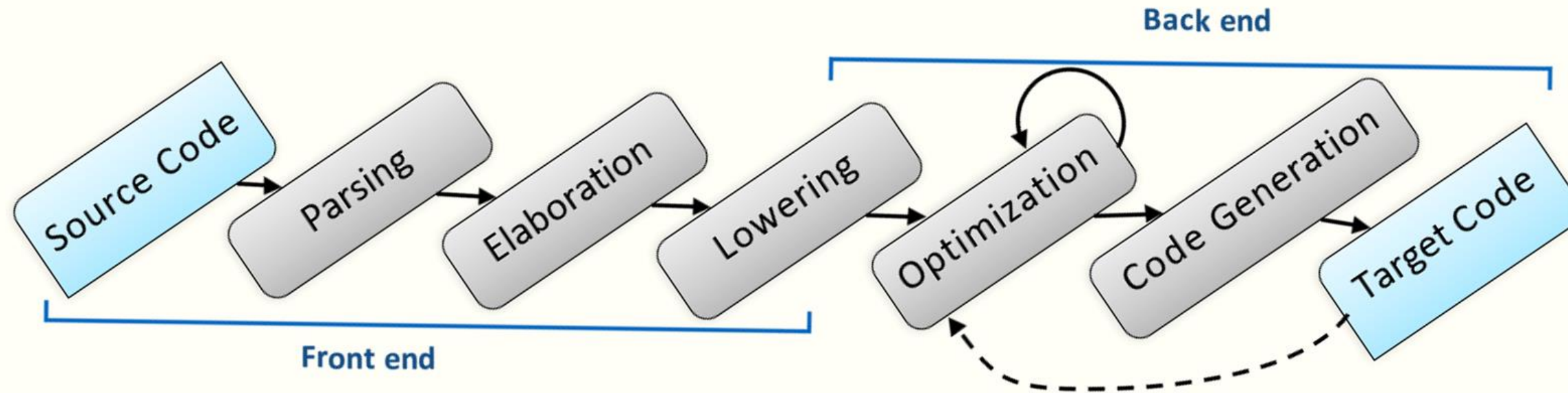
- ❑ How are programs written in a high-level language transformed into machine code?
- ❑ How can we ensure a program is somewhat meaningful?
- ❑ How is the program's memory managed?
- ❑ How can we analyze programs to discover invariant properties and improve their runtime performance?

Historical Note

- ❑ Until the 1950's: computers were programmed in assembly.
- ❑ 1951–1952: **Grace Hopper** developed the A-0 system for the UNIVAC I
 - She later contributed significantly to the design of COBOL
- ❑ 1957: the FORTRAN compiler was built at IBM
 - Team led by John Backus
- ❑ 1960's: development of the first bootstrapping compiler for LISP
- ❑ 1970's: language/compiler design blossomed
- ❑ Today: thousands of languages (most little used)
 - Some better designed than others...

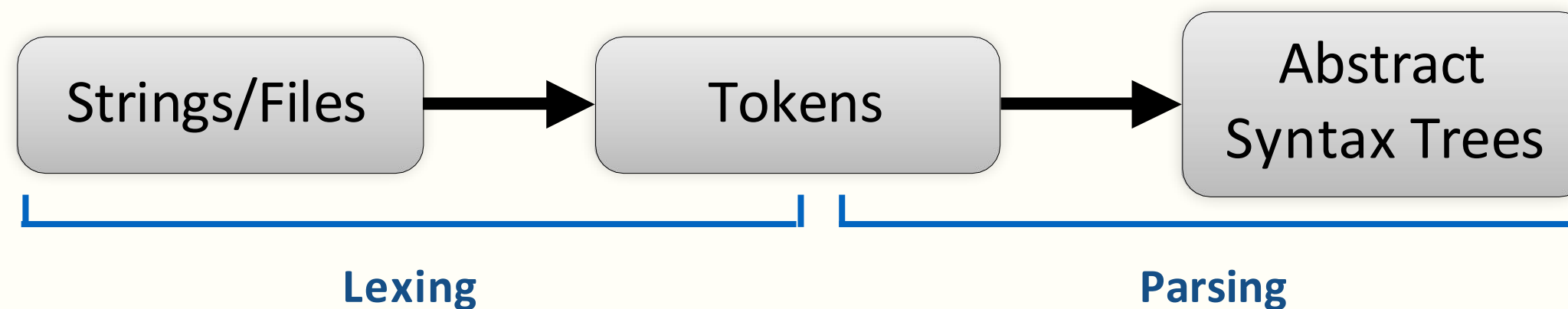


Compiler Architecture



❑ Front end: Lexing & Parsing

- From strings to data structures
- Usually split into two phases:



Parsing tools

- ❑ Parsing is something that happens in essentially all applications.
 - E.g., Google search bar, calendar, etc.
 - First step in going from raw data to information
- ❑ Lots of CS theory to help out
 - Regular expressions (finite state automata)
 - Context-free grammars (push-down automata)
 - These abstractions show up in optimization too!
- ❑ Lots of tool support
 - E.g., Lex and Yacc, Menhir, Antlr, parsing combinators, etc.

Elaboration & Lowering

❑ Elaboration: mainly type-checking

- Check that operations are given values of the right types
- Resolve variables, modules, etc.
- Infer types for sub-expressions
- Check for other safety/security problems

Elaboration

Untyped Abstract
Syntax Trees



Typed Abstract
Syntax Trees

❑ Lowering: translate high-level features into a small number of target-like constructs

- e.g., while, for, do-loops all compiled to code using goto's
- e.g., objects, closures to records and function pointers
- e.g., make type-tests, array-bounds checks, etc. explicit

Lowering



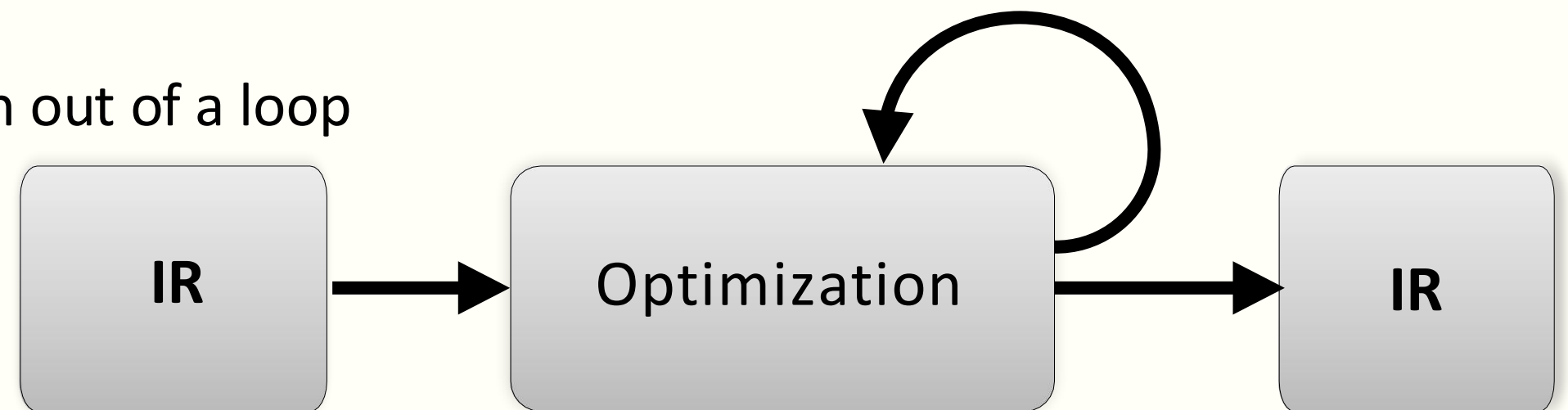
Intermediate Code
(IR)

Optimization & Code generation

❑ Optimization:

- Rewrite expensive sequences of operations into less expensive

- e.g., constant folding: $3+4 \rightarrow 7$
- e.g., lift an invariant computation out of a loop
- e.g., parallelize a loop



❑ Code generation:

- Translate intermediate code into target code

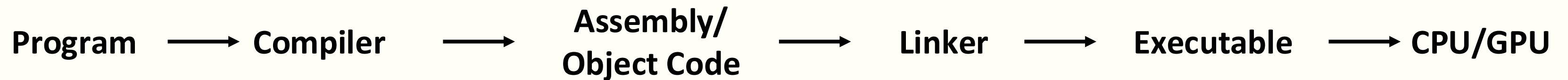
- Register assignment
- Instruction selection
- Instruction scheduling
- Machine-specific optimization



Compiling vs. Interpretation

❑ Compiler translates a program

● C/C++, Rust, Go, Fortran,



❑ Interpreter executes a program

● Python, Ruby, JavaScript, PHP,



❑ Modern Implementation often combine both

● Java, C#, Python(Cpython), Ocaml,



Example: compile, assemble, link, interpret

File: a.ml

```
let add x y = x + y
```

File: main.ml

```
let () = let r = A.add 3 4 in  
Printf.printf "3 + 4 = %d\n" r
```

Compile, Assemble, Link, and Execute

```
# ocamlpt -S -c a.ml  
// produce a.cmi, a.cmx, a.s, a.o  
# ocamlpt -S -c main.ml  
// produce main.[cmi|cmx|s|o]  
# ocamlpt -o main a.cmx main.cmx  
// link and produce the executable  
# ./main  
// output 3 + 4 = 7
```

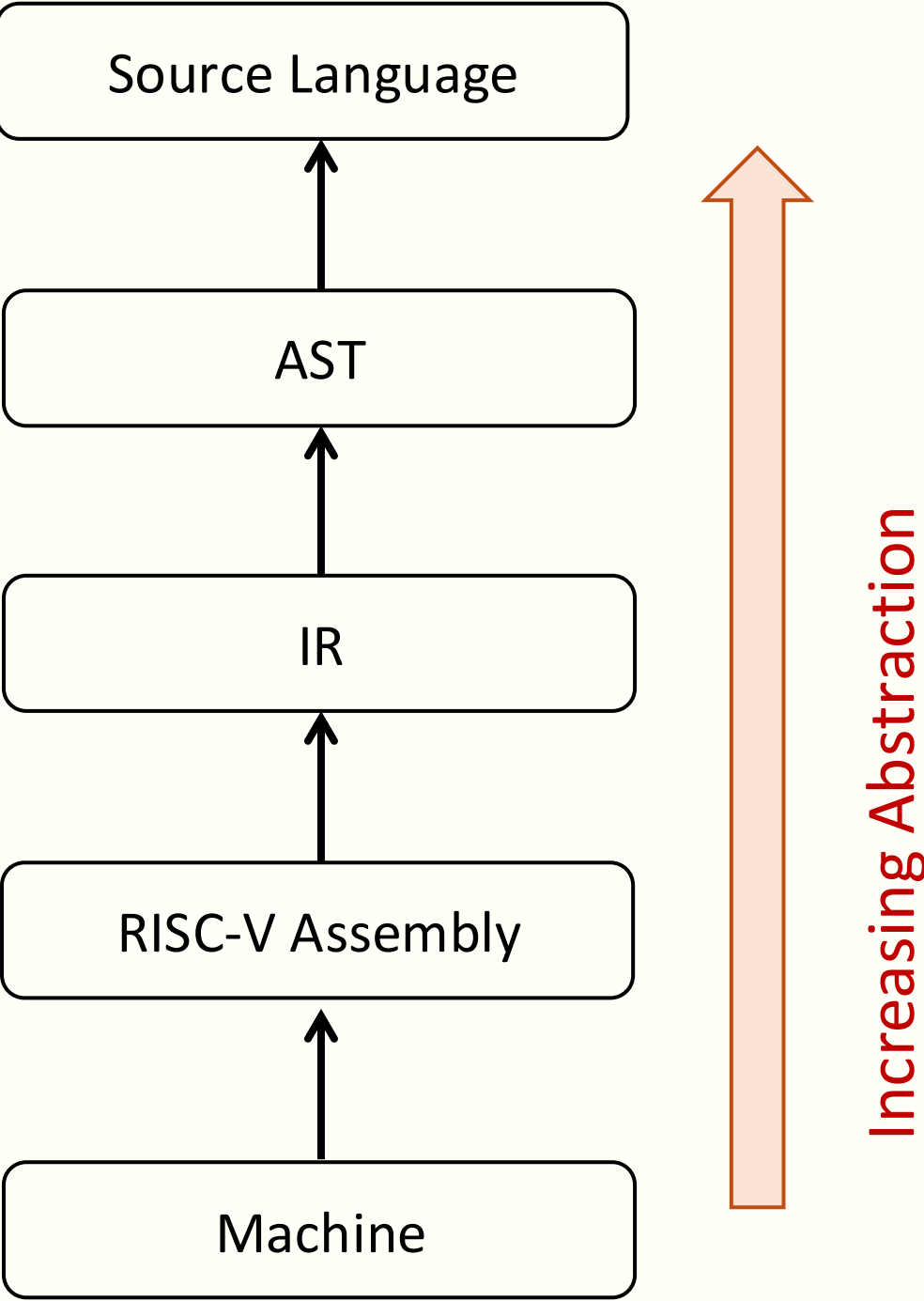
Compile, Link, and Interpret

```
# ocamlc -c a.ml  
// produce a.cmi, a.cmo  
# ocamlc -c main.ml  
// produce main.cmi, main.cmo  
# ocamlc -o main a.cmo main.cmo  
// link and produce the executable  
# ocamlrun main  
// interpret and output 3 + 4 = 7
```

Course Design

#	课程内容	核心内容
1	OCaml 简介	函数式编程、Pattern Matching、ADT、模块
2	IR 与解释执行	IR、基本块、控制流、Interpreter
3	RISC-V 汇编、调用约定与链接	指令、寄存器、栈帧、Calling Convention、Linking
4	指令选择	IR → Machine Instructions、Tree Matching
5	寄存器分配	Virtual/Physical Registers、Liveness、Graph Coloring
6	编译器优化	CFG、数据流分析、Constant Propagation、DCE 等
7	运行时系统与垃圾收集	Stack、Heap、Allocation、GC
8	抽象语法与 AST	Expressions、Statements、AST、Pattern Matching
9	语义分析：作用域与符号表	Scope、Name Resolution、Symbol Table
10	类型检查	Type System、Type Checking、Type Errors
11	IR 生成：AST → IR	Lowering、Control Flow、Temporary Values
12	语法分析	CFG、Recursive Descent、LL、LR
13	词法分析	Regular Expression、NFA、DFA、Lexer
14	面向对象语言支持	Object Layout、Inheritance、Dynamic Dispatch
15	函数式语言支持	First-class Functions、Closure、Environment
16	现代编译技术概览	SSA、JIT、LLVM/MLIR、GC、并行/GPU、AI 编译等

The Abstraction Ladder



Who should take this course?

- ❑ People fascinated by languages & compilers
 - Why does[n't] this language include this feature?
 - Systems programmers
- ❑ Know the one tool that you use every day.
 - What can[t] a compiler do well?
- ❑ Architecture designers
 - Interdependence between compiler and architecture
 - See **Intel iAPX 432** and **Intel Itanium** (compiler-related) failures; **register windows**
 - These days, architecture folks use compilers too!
- ❑ API designers
 - A language is the ultimate API
 - c.f., Facebook's Hack language

Learning outcomes

□ You will learn:

- Practical applications of theory
- Lexing/Parsing/Interpreters
- How high-level languages are implemented in machine language
- (A subset of) RISC-V architecture
- More about common compilation tools like GCC and LLVM
- A deeper understanding of code
- A little about programming language semantics, program analysis, & types
- How to manipulate complex data structures
- How to be a better programmer

Assessment

□ Roughly:

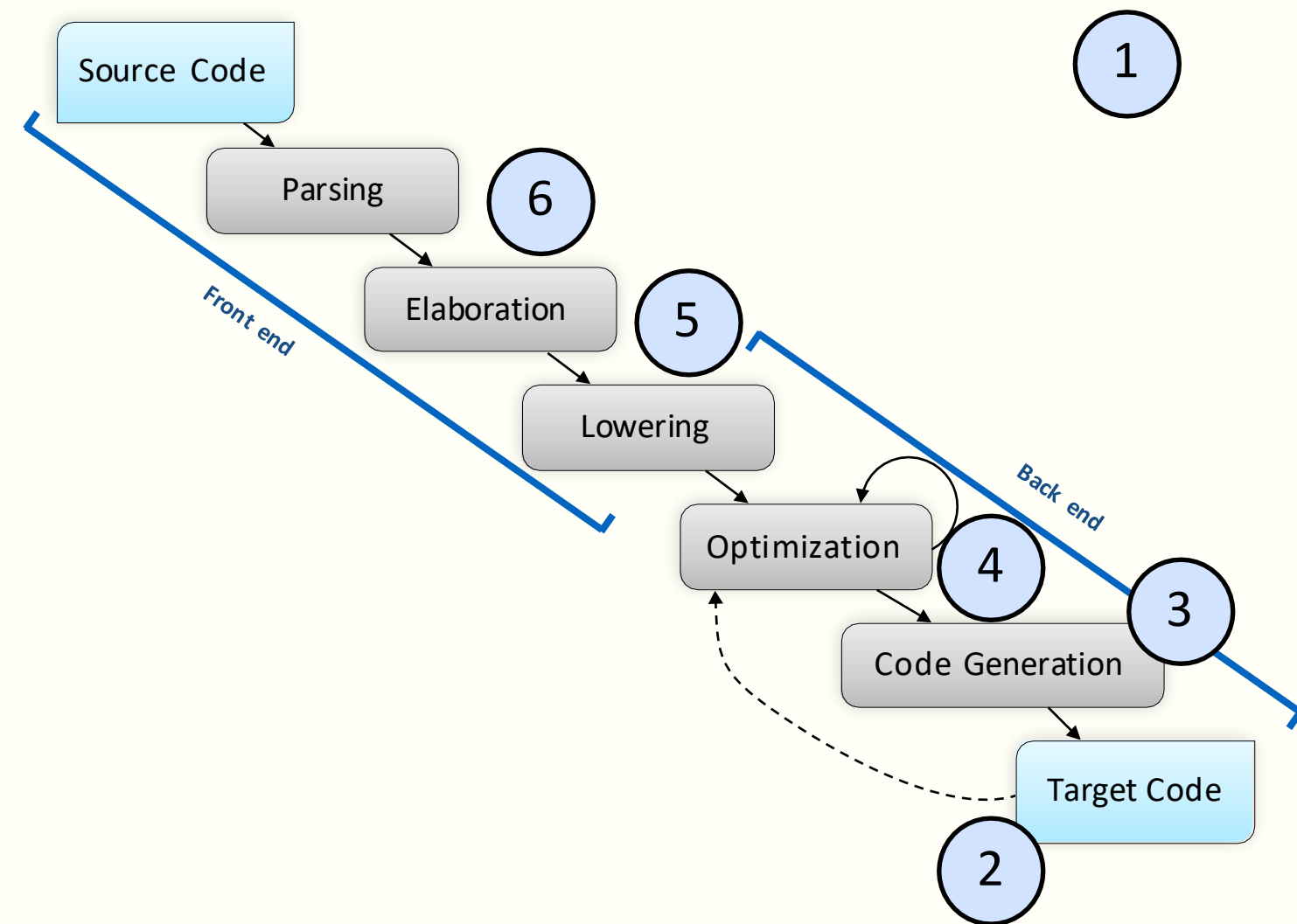
- 60% final exam
- 30% projects (~6 projects total, not equally weighted)
 - Submit them online
- 10% participation and assignments
 - E.g., attend lectures and engage in discussion, finish assignments, ...
 - 要求所有学生全勤，感觉人数不够63时，会根据点名册把缺勤学生找出来

Projects

- ❑ 6 projects, mostly implementing parts of a compiler
- ❑ Typically 2-3 weeks per project
- ❑ Programming environment: in **Ocaml**  <https://ocaml.org/>
 - Ideal for writing compilers!
- ❑ Implementation heavy course!
 - All submissions must type-check and compile
 - Multiple projects out at same time
- ❑ Online Judge:
 - <http://47.109.176.53:8088> (<http://compiler.plc-cqu.cn> 备案中)

Projects

- ❑ 1. HelloCaml
 - Refresh OCaml coding
 - Implement interpreter and Understand IR
- ❑ 2. RISC-V
 - RISC-V Assembler and Linker
 - Understand the machine we are targeting
- ❑ 3. Code Generation
 - Compile IR to RISC-V
 - Garbage collection and Register allocation
- ❑ 4. Dataflow analysis and optimizations
- ❑ 5. AST
 - Type checking
 - Simple lowering: from AST to IR
- ❑ 6. Lang: our language
 - Simple front-end: Lexing and parsing
 - Structs, function pointers, Classes



作业与实验说明

□作业

- 独立完成，电子形式（PDF）提交到助教邮箱278487953@qq.com
- 允许纸上完成后拍照或扫描上传，需确保内容完整、字迹清晰
- 截止时间：发布后一周内提交（100%），两周(60%)，三周（30%），其他0%
- 作业反馈：发布后第二周公布参考答案

□实验

- 独立完成，OJ上提交，自动评测（最多允许提交10次，取最高分）
- 允许讨论，咨询AI工具，但禁止分享代码
- 截止时间：发布后三周内，逾期提交不计分

Collaboration/Academic integrity

- ❑ Projects are done individually: you must write all your own code.
 - Don't share your code with others
 - Don't accept code from others
 - Don't look for solutions online
 - **Don't post course materials to websites or course-content archives**
 - Make sure your GitHub repo is private!
- ❑ But: study groups are a very effective way to learn from your peers
- ❑ Guidelines for working with others:
 - OK to talk about high-level ideas: understanding concepts, how to approach an implementation
 - Use words, not code
 - OK to talk about low-level details
 - How to use an OCaml library, etc.
- ❑ If you are ever in doubt, ask the course staff to clarify what is and isn't appropriate.

References

□ We build this course by primarily referencing the following materials:

- CS 153: Compilers, Harvard University, by Prof. Stephen Chong

- <https://groups.seas.harvard.edu/courses/cs153/2019fa/schedule.html>

- Compiler Construction, University of Cambridge, by Jeremy Yallop

- <https://www.cl.cam.ac.uk/teaching/2324/CompConstr/materials.html>

- CS 143: Compilers, Stanford University, by Keith Schwarz

- <https://web.stanford.edu/class/archive/cs/cs143/cs143.1128/>

Questions/comments?



群聊：编译原理课程群

- 用于即时/离线交流
- 请各位同学使用“学号-姓名”格式作为名片/昵称



该二维码7天内(9月13日前)有效，重新进入将更新